Optimized Placement and Pricing Framework for Electric Vehicle Charging Stations

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By

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DECLARATION

I hereby certify that the work presented in this report/thesis, entitled **"Optimized Placement and Pricing Framework for Electric Vehicle Charging Stations"**, submitted in partial fulfillment of the requirements for the Integrated Post Graduate - Master of Technology in Information Technology, is an authentic record of my own work carried out during the period **July-2025 to March-2026** under the supervision of **Dr. Aditya Trivedi** and **Dr. Saumya Bhadauria**. All references, figures, and tables taken from other sources have been properly cited.

Dated: **Signature of the candidate**

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Dated: **Signature of Supervisor: Prof. Aditya Trivedi**

Dated: **Signature of Supervisor Dr. Saumya Bhadauria**

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Ankit Mewada

Abstract

This thesis presents a four-objective optimization framework for the strategic placement of Electric Vehicle Charging Stations (EVCS) across the 85 wards of Indore, Madhya Pradesh. The framework simultaneously minimizes total capital expenditure and mean queue waiting time while maximizing spatial demand coverage and net return on investment — objectives that are inherently conflicting and require a Pareto-based solution approach. Demand is modelled using a logistic EV adoption curve (inflection at 2030) anchored to ward-level CAGR-based demographic projections, supplemented by 100-scenario Monte Carlo stochastic simulation. Service quality is captured by an M/M/c/K finite-capacity queueing model evaluated across three Time-of-Use demand periods, with a hard blocking probability constraint ($P_{\text{block}} \leq 10\%$) enforced via a penalty term in the optimization loop whenever the constraint is violated.

Two evolutionary algorithms — Non-dominated Sorting Genetic Algorithm II (NSGA-II) and Multi-Objective Particle Swarm Optimization (MOPSO) — are applied to a heterogeneous pool of 30+ candidate venues across five functional zones, with zone-adjusted service radii following MoRTH guidelines. Performance is formally evaluated using Hypervolume (HV), Generational Distance (GD), and Spread indicators. NSGA-II (HV = 0.8412, GD = 0.0014, median $W_q =$ 3.9 min) outperforms MOPSO (HV = 0.7205, GD = 0.0039, median $W_q =$ 52.2 min) across all solution-quality and service metrics. The recommended NSGA-II optimal solution deploys 24 stations, achieves 86.4% city-wide demand coverage, maintains a mean queue waiting time of 4.8 minutes, and delivers a net annual ROI of 14.2% for a total capital outlay of 2.14 Crore. Sensitivity analysis confirms robustness to EV growth rate and electricity tariff variations, and scalability benchmarks demonstrate polynomial runtime scaling suitable for larger metropolitan deployments.

Contents

List of Figures

List of Tables

List of Acronyms

BEE	Bureau of Energy Efficiency
CAGR	Compound Annual Growth Rate
CAPEX	Capital Expenditure
CSV	Comma Separated Values
DEAP	Distributed Evolutionary Algorithms in Python
EV	Electric Vehicle
EVCS	Electric Vehicle Charging Station
EVCN	Electric Vehicle Charging Network
EVSE	Electric Vehicle Supply Equipment
FAME	Faster Adoption and Manufacturing of Hybrid and Electric Vehicles
FCFS	First-Come First-Served (Queue Discipline)
GA	Genetic Algorithm
GD	Generational Distance
GIS	Geographic Information System
HV	Hypervolume Indicator
ILP	Integer Linear Programming
IMC	Indore Municipal Corporation
INR	Indian National Rupee
M/M/c	Multi-Server Infinite-Capacity Queueing Model
M/M/c/K	Multi-Server Finite-Capacity Queueing Model
MINLP	Mixed-Integer Nonlinear Programming

MOEA	Multi-Objective Evolutionary Algorithm
MOO	Multi-Objective Optimization
MOPSO	Multi-Objective Particle Swarm Optimization
MoRTH	Ministry of Road Transport and Highways
NEMMP	National Electric Mobility Mission Plan
NSGA-II	Non-Dominated Sorting Genetic Algorithm–II
OPEX	Operational Expenditure
OSM	OpenStreetMap
PASTA	Poisson Arrivals See Time Averages
PLI	Production Linked Incentive
PPP	Public-Private Partnership
PSO	Particle Swarm Optimisation
QoS	Quality of Service
ROI	Return on Investment
SOC	State of Charge (EV Battery)
SP	Subproblem (Benders Decomposition)
SPP	Station Placement Problem
TCO	Total Cost of Ownership
ToU	Time-of-Use
ULF	Utilization Load Factor
V2G	Vehicle-to-Grid
W_q	Average Waiting Time in Queue

1

Introduction

This chapter introduces the motivation and background for developing a data-driven, multi-objective optimization framework for large-scale Electric Vehicle Charging Station (EVCS) planning in rapidly growing urban environments. Section 1.1 provides a general introduction to the research context, emphasizing the increasing importance of optimized EVCS networks for sustainable mobility in cities such as Indore. Section 1.2 outlines the technical background—covering EV adoption trends, spatial demand characteristics, grid constraints, and the rationale for integrating NSGA-II, Benders decomposition, CAGR-based demand forecasting, and queueing models in EVCS planning. Section 1.3 highlights the practical and methodological motivations guiding the work, focusing on the need for scalable, data-driven approaches that balance cost, coverage, service quality, and profitability. Section 1.4 lists the primary objectives of the thesis, including optimal station placement, dynamic pricing optimization, and operational performance evaluation. Finally, Section 1.5 presents the overall layout of the report, summarizing the structure and interconnections of the subsequent chapters.

1.1 Introduction

The rapid proliferation of Electric Vehicles (EVs) in Indian cities has created an urgent need for scientifically planned public charging infrastructure. Inadequately placed stations lead to range anxiety, long queuing times, and underutilised investments — outcomes that collectively erode user confidence and slow the broader EV adoption transition. The city of Indore, a major commercial hub of Madhya Pradesh, presents a challenging yet representative context for this problem, given its diverse ward-level demography, heterogeneous land-use patterns, and an active municipal push toward sustainable urban mobility under the Smart Cities Mission. As of early 2024, the city had fewer than 50 publicly accessible EV charging points — a significant deficit relative to its projected EV fleet size under any realistic adoption scenario.

Designing an optimized EVCS network is fundamentally a combinatorial multi-objective problem. A planner must simultaneously minimize capital expenditure, maximize spatial demand coverage, maximize financial return on investment, and minimize customer queue waiting times — objectives that are inherently conflicting. No single solution dominates all others across every objective; instead, a Pareto-optimal set of trade-off solutions exists, which is the domain of multi-objective evolutionary algorithms (MOEAs). This thesis addresses this problem through a comprehensive four-objective optimization framework applied across the 85 administrative wards of Indore, comparing two state-of-the-art evolutionary algorithms — Non-dominated Sorting Genetic Algorithm II (NSGA-II) and Multi-Objective Particle Swarm Optimization (MOPSO) — and validating results through analytical queueing models and Monte Carlo stochastic demand simulation.

1.1.1 Overview of the Proposed Framework

The global transition toward sustainable transportation has led to a rapid increase in EV adoption. In India, and particularly in fast-growing cities such as Indore, this trend is reinforced by government incentives under the FAME II scheme, environmental regulations, and rising public awareness. However, large-scale EV uptake is fundamentally constrained by the availability, accessibility, and reliability of the charging infrastructure. Poorly planned EVCS networks can lead to range anxiety, long waiting times, uneven spatial coverage, and overloading of the power grid.

Designing an optimized EVCS network is a complex, multi-objective task. It involves simultaneously deciding:

- which candidate locations should host charging stations;
- how many chargers and what capacity should be installed at each selected site;
- how charging prices should be set across the network under Time-of-Use (ToU) demand patterns;
- how to ensure that grid constraints, blocking probability thresholds, and user service quality requirements are simultaneously satisfied.

These decisions must be made under spatially heterogeneous demand, limited grid capacity, stochastic demand uncertainty, and evolving population and EV growth over a multi-year horizon.

To address this challenge, the thesis proposes a unified framework with five tightly coupled components:

(i) **Four-objective problem formulation:** The EVCS placement problem is formulated as a binary combinatorial optimisation problem minimizing total capital expenditure (f_1) and mean queue waiting time (f_4), while maximizing spatial demand coverage (f_2) and net return on investment (f_3), over a pool of 30+ heterogeneous candidate venues across the 85 wards of Indore.

(ii) **NSGA-II and MOPSO comparative optimization:** Both algorithms are independently applied to generate Pareto-optimal station deployment plans. NSGA-II employs fast non-dominated sorting with crowding distance selection, while MOPSO uses an external archive with crowding-based leader selection. Algorithm quality is formally compared using Hypervolume (HV), Generational Distance (GD), and Spread performance indicators.

(iii) **Monte Carlo stochastic demand modelling:** Demand is projected using a logistic EV adoption curve anchored to CAGR-based ward-level demographic forecasts, and further overlaid with 100-scenario Monte Carlo simulation (noise fraction $v = 0.15$) to ensure that selected station configurations are robust to stochastic demand fluctuations.

(iv) **M/M/c/K queueing model with Time-of-Use pricing:** Service quality at each station is captured by an M/M/c/K finite-capacity queueing model with $K = c + 8$ physical waiting bays, evaluated across three ToU demand periods — Peak, Normal, and Idle. A blocking probability constraint ($P_{\text{block}} > 10\%$) is embedded directly into the evolutionary evaluation loop as a hard penalty, ensuring that no selected station configuration becomes queue-infeasible under peak demand.

(v) **Sensitivity analysis and scalability benchmarking:** The robustness of the optimal solutions is examined with respect to EV growth rate, electricity tariff, installation cost factor, and charger service rate. Scalability benchmarks confirm polynomial runtime growth for both algorithms, validating applicability to larger metropolitan deployments.

Applied to Indore, the framework produces a recommended NSGA-II optimal solution that deploys 24 stations, achieves 86.4% city-wide spatial demand coverage, maintains a mean queue waiting time of 4.8 minutes, and delivers a net annual ROI of 14.2%, for a total capital outlay of 2.14 Crore. NSGA-II achieves a Hypervolume of 0.8412 and a median waiting time of 3.9 minutes across its Pareto front, outperforming MOPSO on all solution-quality and service-reliability metrics.

1.2 Background

The expansion of EV infrastructure in Indian cities requires optimization techniques capable of handling multiple layers of complexity. Unlike conventional fuel stations, EVCSs interact directly with the power distribution network, making their planning and operation depend simultaneously on transportation patterns, demographic trends, grid constraints, and financial viability.

1.2.1 EV Adoption and Urban Charging Needs

India ranks as the world's third largest automobile market and has set an ambitious target of 30% EV penetration by 2030 under the National Electric Mobility Mission Plan (NEMMP). As of 2024, India had registered over four million EVs, with two-wheelers constituting the dominant segment. For Indore specifically, the population grew from approximately 1.87 million in 2011 to an estimated 3.67 million in 2021, corresponding to a CAGR of roughly 7.04%, with projections suggesting the population may exceed 7.2 million by 2031. Under a moderate EV adoption scenario, the EV fleet is expected to increase by more than a factor of eight over the same period.

Such growth patterns imply that EVCS networks cannot be designed solely for current demand. They must be robust to substantial increases in both the number of EVs and their spatial distribution, motivating the use of logistic adoption curves and multi-scenario stochastic demand models rather than static planning assumptions.

1.2.2 Limitations of Traditional Planning Approaches

Traditional EVCS planning approaches often rely on:

- single-objective deterministic optimization (e.g., minimizing average distance or total cost alone);
- simplified queueing assumptions such as the Erlang-B loss model, which ignores physical waiting bays and overestimates blocking;

- static demand assumptions that do not account for long-term EV adoption growth or stochastic variability;
- uniform service radii that ignore urban density heterogeneity and road-network accessibility differences.

These models are insufficient for city-scale planning where multiple conflicting objectives must be balanced simultaneously. In particular, they fail to capture trade-offs between cost, coverage, service quality, and financial sustainability — or to enforce service-level constraints such as maximum blocking probability during peak demand periods.

1.2.3 NSGA-II, MOPSO, and Queueing Models

NSGA-II is an elitist multi-objective genetic algorithm that employs fast non-dominated sorting with $\mathcal{O}(MN^2)$ complexity and a crowding distance diversity mechanism. It has become the benchmark algorithm for multi-objective combinatorial optimisation, with well-understood convergence guarantees and binary crossover/mutation operators that map naturally onto station selection problems.

MOPSO extends the Particle Swarm Optimisation paradigm to multiple objectives through an external archive of non-dominated solutions and crowding-based leader selection. For binary problems, continuous particle positions are binarised via a threshold for objective evaluation, while velocity updates operate in continuous space. MOPSO is known to converge faster than NSGA-II in early iterations, motivating its inclusion as a comparative benchmark.

The M/M/c/K finite-capacity queueing model provides analytical steady-state expressions for queue length L_q, waiting time W_q, server utilisation ρ, and blocking probability P_{block}. The finite buffer assumption ($K = c + W$ waiting bays) is more realistic for urban charging stations than infinite-queue M/M/c models, and Time-of-Use segmentation of the arrival rate captures the bimodal daily demand pattern observed in field data.

1.3 Motivation

The motivation for this research arises from both practical infrastructure requirements and methodological gaps in existing EVCS planning literature.

1. Addressing Infrastructure Deficits in Indian Cities

Most multi-objective EVCS placement studies have been conducted for Chinese or European cities using generic urban models. Studies specifically for Tier-1 Indian cities with real ward-level demographic data, India-specific policy parameters, and heterogeneous venue types are scarce. Indore, with its 85 wards, diverse land-use, and active smart-city programme, provides an empirically grounded and practically relevant case study.

2. Integrated Four-Objective Formulation

Existing studies typically address two or three objectives. The simultaneous optimisation of capital expenditure, spatial coverage, net ROI, and queue waiting time — with net ROI defined as annual profit divided by CAPEX, rather than raw profit, to avoid degeneracy with the coverage objective — has not been reported for Indian cities. This formulation creates genuine trade-off diversity in the Pareto search space.

3. Realistic Queueing with Blocking Constraints

Prior work either uses over-simplified Erlang-B loss models or time-averaged M/M/c queues. An integrated M/M/c/K model with ToU demand periods and a blocking probability hard constraint embedded inside the evolutionary evaluation loop represents a methodological advance that forces the optimizer to provision sufficient charger capacity at demand bottlenecks.

4. Stochastic Demand Robustness

Deterministic demand forecasts are insufficient for infrastructure planning over a 10-year horizon. Monte Carlo simulation with 100 demand scenarios ensures that selected station configurations remain feasible under the full range of plausible future demand trajectories, not merely under the mean forecast.

5. Reproducible, Extensible Planning Tool

The combination of open demographic data, a modular Python implementation, two well-documented evolutionary algorithms, and analytical queueing models yields a framework that is reproducible, extensible to other cities, and capable of accommodating future enhancements such as renewable integration, vehicle-to-grid (V2G) capabilities, and real-time dynamic pricing.

1.4 Thesis Objectives

The specific objectives of this thesis are:

(i) To develop a data-driven multi-objective optimization model for EVCS placement across Indore's 85 wards, integrating a CAGR-based logistic EV adoption curve with Monte Carlo uncertainty quantification to forecast ward-level charging demand, and formulating the station selection problem as a four-objective binary optimisation incorporating total capital expenditure, spatial demand coverage, net return on investment, and mean queue waiting time.

(ii) To implement and comparatively evaluate NSGA-II and MOPSO on the formulated problem, embedding an analytical M/M/c/K finite-capacity queueing model with Time-of-Use demand periods and a hard blocking probability constraint ($P_{\text{block}} \leq 10\%$) directly within the evolutionary evaluation loop, and quantifying algorithm quality using Hypervolume (HV), Generational Distance (GD), and Spread performance indicators.

(iii) To validate the robustness and scalability of the proposed framework through sensitivity analysis with respect to EV growth rate, electricity tariff, installation cost, and charger service rate, and through scalability benchmarks confirming polynomial runtime growth suitable for larger metropolitan deployments.

1.5 Layout of the Thesis

This report is structured into the following chapters:

- **Chapter 2: Literature Review** — Reviews existing work on EVCS placement, multi-objective evolutionary algorithms, M/M/c/K queueing models, and stochastic demand forecasting methods. It identifies five research gaps that motivate the proposed framework and provides a structured comparative summary of twelve representative studies.
- **Chapter 3: Model Architecture** — Describes the complete architecture of the proposed framework, including the deterministic baseline model, four-objective problem formulation, NSGA-II and MOPSO optimization layers, M/M/c/K queueing evaluation layer with Time-of-Use demand segmentation, and formal algorithm performance evaluation using HV, GD, and Spread indicators.
- **Chapter 4: Methodology** — Details the structured multi-stage pipeline covering dataset acquisition, spatial zoning, ward-level logistic demand forecasting, Monte Carlo stochastic simulation, four-objective optimization using NSGA-II and MOPSO, queueing-based operational evaluation, sensitivity analysis, and scalability benchmarking.
- **Chapter 5: Results and Analysis** — Presents Pareto front comparisons between NSGA-II and MOPSO, formal algorithm performance indicators, optimal solution analysis, queue waiting time distributions, HV convergence behaviour, sensitivity studies on EV growth rate and tariff parameters, and scalability benchmarks.
- **Chapter 6: Conclusion and Future Work** — Summarizes the main contributions, states deployment recommendations for Indore, and outlines future extensions

including road-network routing, heterogeneous charger-type optimisation, battery swapping station integration, and statistical significance testing.

2

Literature Review

Section 2.1 reviews twelve influential studies spanning EVCS siting, multi-objective evolutionary algorithms, M/M/c/K queueing models with Time-of-Use demand segmentation, stochastic demand forecasting, and formal NSGA-II vs. MOPSO comparative evaluation. Together, these works establish the theoretical and methodological foundation for the four-objective optimization framework proposed in this thesis. Section 2.2 identifies five key research gaps — the absence of Indian city studies with real ward-level data, lack of integrated four-objective formulations, over-simplified queueing models applied post-hoc rather than embedded in the optimization loop, absence of formal algorithm comparison on binary EVCS placement problems, and use of raw profit as a financially degenerate objective — that collectively motivate the proposed framework. Section 2.3 provides a structured comparative summary of the twelve reviewed studies, consolidating each study's methodology, key contribution, and limitation to offer a clear overview of existing approaches and the gaps addressed by this thesis.

2.1 Literature Survey

This section reviews influential studies relevant to electric vehicle charging station (EVCS) planning, multi-objective evolutionary algorithms, queueing-based service modelling, stochastic demand forecasting, and comparative algorithm evaluation. These works collectively shape the conceptual and methodological foundation of the four-objective NSGA-II and MOPSO framework proposed in this thesis.

(i) **Zhao et al. (2023)** propose a bi-level EVCS location model that accounts for behavioural diversity in user charging modes (self-charging and valet-charging). Their robust Monte Carlo demand integration demonstrates that ignoring demand uncertainty leads to over-optimistic coverage estimates, directly informing the 100-scenario stochastic simulation module adopted in this work. Pricing, grid capacity, and queueing constraints remain absent from their framework.

(ii) **Deb et al. (2002)** introduce NSGA-II, the benchmark elitist multi-objective genetic algorithm employing fast non-dominated sorting with $\mathcal{O}(MN^2)$ complexity and crowding distance diversity preservation. Its elitism, well-understood convergence properties, and compatibility with binary decision variables make it the primary optimizer in this thesis. However, it does not natively incorporate queueing or pricing constraints within the evaluation loop.

(iii) **Coello Coello and Lechuga (2002)** propose MOPSO, extending Particle Swarm Optimisation to multiple objectives through an external archive of non-dominated solutions and crowding-based leader selection. MOPSO is known to converge faster than NSGA-II in early iterations, motivating its inclusion as a comparative benchmark in this work. Its weakness for binary combinatorial problems — where continuous velocity mechanics must be binarised via a threshold — is examined in the results chapter.

(iv) **Paul and Bhattacharya (2021)** study multi-objective EVCS siting for Kolkata using NSGA-II with cost and coverage objectives, demonstrating the superiority of evolutionary approaches over classical programming for large-scale Indian city instances. Their work is restricted to two objectives and does not incorporate queueing, ROI, or stochastic demand — gaps that directly motivate the four-objective formulation of this thesis.

(v) **Sadeghi-Barzani et al. (2014)** optimise the placement and sizing of fast charging stations by minimising installation cost and power loss under grid-side constraints, providing early insights into power-system-aware EVCS siting. The work remains single-objective and does not consider user accessibility, queue performance, or financial viability — limitations that motivate the multi-objective formulation herein.

(vi) **Bernardo et al. (2021)** demonstrate that urban density significantly affects the appropriate service radius, with dense city centres requiring smaller radii (∼1–2 km) than suburban areas (∼3–5 km). This finding directly informs the zone-adjusted service radii of 1.5 km (Central Core) and 2.5 km (Outer zones) adopted in this thesis following MoRTH guidelines.

(vii) **Tan et al. (2016)** introduce a three-period Time-of-Use (ToU) arrival rate model for EV charging queues — Peak, Normal, and Idle — finding that this segmentation accurately captures the bimodal daily demand pattern observed in field data. Their framework is extended in this thesis by embedding the ToU-weighted M/M/c/K evaluation inside the evolutionary loop rather than applying it as a post-hoc analysis.

(viii) **Zhang et al. (2019)** demonstrate that ignoring finite system capacity in queueing models leads to underestimation of blocking probability by up to 15–20% at high utilization rates, justifying the use of the M/M/c/K model with $K = c+W$ physical waiting bays over the simpler Erlang-B loss model. Their capacity planning framework for fast charging stations forms the analytical basis of the queueing evaluation layer in this work.

(ix) **Li et al. (2022)** jointly optimise charging station placement and pricing under stochastic arrival using chance-constrained programming. Their work highlights the importance of treating pricing as an integrated objective rather than a post-hoc decision. In this thesis, net ROI — defined as annual profit divided by CAPEX — replaces raw profit as an independent objective to avoid near-perfect linear correlation with coverage that would degenerate the Pareto front.

(x) **Reyes-Sierra and Coello Coello (2006)** conduct a comprehensive comparative survey of NSGA-II and MOPSO across multiple problem domains, finding that MOPSO converges faster in early iterations but sacrifices spread diversity, while NSGA-II maintains a better-distributed Pareto front. For binary combinatorial placement problems, NSGA-II has been reported to outperform MOPSO in hypervolume indicators. This theoretical prediction is directly verified in the results of this thesis through formal HV, GD, and Spread evaluation.

(xi) **Xia et al. (2022)** show that ignoring power network constraints in EV charging station placement leads to infeasible solutions at high EV penetration rates, and propose penalising solutions that exceed zone-level substation capacity. This per-zone grid capacity constraint is adopted explicitly in the problem formulation of this thesis.

(xii) **Kulkarni et al. (2021)** apply the generalised Bass logistic diffusion model to forecast EV adoption in India, estimating an inflection point near 2030 consistent with NITI Aayog's aspirational policy targets and battery cost parity trajectories. Their parameterisation ($t_0 = 2030$, $a = 0.3$) is adopted in the logistic adoption curve of this thesis and validated against the ward-level demand projections generated from Census 2011 data.

2.2 Research Gaps

The review above reveals five methodological gaps that collectively motivate the proposed framework:

(i) **Indian city focus with real ward-level data.** Most multi-objective EVCS studies use Chinese or European city models. Studies for Tier-1 Indian cities with actual ward-level demographic data and India-specific policy parameters are scarce.

(ii) **Integrated four-objective formulation.** Existing studies address two or three objectives. The simultaneous optimisation of capital expenditure, spatial coverage, net ROI, and queue waiting time in a single Pareto framework has not been reported for Indian cities.

(iii) **M/M/c/K with ToU pricing embedded in the optimization loop.** Prior work either uses over-simplified Erlang-B models or applies time-averaged M/M/c queues post-hoc. An M/M/c/K model with ToU demand periods and a hard blocking probability constraint ($P_{\text{block}} \leq 10\%$) embedded directly inside the evolutionary evaluation represents a methodological advance.

(iv) **NSGA-II vs. MOPSO formal comparison on binary placement problems.** While comparative surveys exist, formal empirical comparison of NSGA-II and MOPSO using HV, GD, and Spread on a binary EVCS placement instance with queueing constraints has not been reported.

(v) **Net ROI as an independent objective.** Using raw profit creates near-perfect linear correlation with coverage ($r > 0.99$), yielding a degenerate Pareto front. Replacing it with net ROI (annual profit / CAPEX) creates genuine trade-off diversity in the multi-objective search space.

2.3 Comparative Summary of Key Literature

No.	Study	Method / Model	Key Contribution / Limitation
1	Zhao et al. (2023)	Bi-level + Monte Carlo	Robust demand modelling; no pricing or queueing.
2	Deb et al. (2002)	NSGA-II	Benchmark MOEA; no queueing or pricing integration.
3	Coello Coello & Lechuga (2002)	MOPSO	Fast convergence; weaker spread on binary problems.
4	Paul & Bhattacharya (2021)	NSGA-II, Indian city	Two-objective only; no queueing or ROI.
5	Sadeghi-Barzani et al. (2014)	Fast-charging siting + grid	Grid-aware siting; single-objective only.
6	Bernardo et al. (2021)	Density-based radius model	Zone-adjusted radii; no optimization framework.
7	Tan et al. (2016)	ToU queueing model	Time-of-use arrivals; not embedded in optimization.
8	Zhang et al. (2019)	M/M/c/K capacity planning	Finite-buffer blocking; no spatial siting.
9	Li et al. (2022)	Joint placement + pricing	Stochastic pricing; no evolutionary multi-objective.
10	Reyes-Sierra & Coello Coello (2006)	NSGA-II vs. MOPSO survey	NSGA-II superior spread; no placement application.
11	Xia et al. (2022)	Grid-constrained placement	Power-feasible siting; single-objective.
12	Kulkarni et al. (2021)	Bass logistic adoption, India	India-calibrated $t_0 = 2030$; no spatial model.

Table 2.1: Comparative summary of key studies on EVCS siting, queueing, multi-objective optimization, and demand forecasting relevant to this thesis.

3

Model Architecture

This chapter presents the model architecture used in our EVCS optimization framework. Section 3.1 describes the baseline multi-objective structure, including demand modeling, cost functions, and grid constraints. Section 3.2 outlines the NSGA-II architecture used for site-selection optimization, and Section 3.3 explains the Benders decomposition module used for dynamic pricing. Section 3.4 summarizes the queueing-based evaluation layer that validates waiting times and utilization after optimization. Section 3.5 identifies the core components of the proposed framework and how they interact to produce scalable, city-level EVCS deployment plans.

3.1 Model Architecture

This chapter presents the complete architecture of the proposed four-objective Electric Vehicle Charging Station (EVCS) optimization framework designed for Indore city. The framework integrates five tightly coupled components in a unified, data-driven pipeline:

- a logistic demand forecasting and Monte Carlo stochastic simulation module,
- a four-objective binary optimisation layer using NSGA-II and MOPSO,
- an M/M/c/K finite-capacity queueing evaluation layer with Time-of-Use demand segmentation,
- a blocking probability constraint enforcement mechanism, and
- a sensitivity analysis and scalability benchmarking module.

These components interact through shared spatial datasets spanning the 85 administrative wards of Indore, a heterogeneous pool of 30+ candidate venues across five functional zones, and a logistic EV adoption curve anchored to CAGR-based ward-level demographic projections. The result is a scalable framework capable of generating Pareto-optimal EVCS deployment strategies evaluated against formal quality indicators: Hypervolume (HV), Generational Distance (GD), and Spread.

Data Layer: Census 2011 → CAGR → Logistic Adoption Curve → Ward-level Daily Demand

Stochastic Layer: Monte Carlo Simulation (100 scenarios, $v = 0.15$)

Optimization Layer: NSGA-II ∥ MOPSO → Pareto-Optimal Station Subsets

Evaluation Layer: M/M/c/K Queue (ToU) + P_{block} Penalty + HV / GD / Spread

Output: Convergence Plots + Pareto Fronts + Sensitivity Analysis + Deployment Recommendation

3.2 Baseline Model

The baseline model provides a deterministic, single-period, cost–coverage formulation used as a reference benchmark for evaluating the multi-objective optimization results. It focuses exclusively on minimizing infrastructure cost while ensuring citywide spatial coverage under grid-capacity constraints.

3.2.1 Deterministic Cost–Coverage Formulation

Let

$$\begin{aligned} \mathcal{W} &: \text{ set of demand wards,} & |\mathcal{W}| &= 85, \\ \mathcal{J} &: \text{ set of candidate sites,} & |\mathcal{J}| &= 30+, \\ \mathcal{Z} &: \text{ set of functional zones,} & |\mathcal{Z}| &= 5. \end{aligned}$$

Decision variable:

$$x_j = \begin{cases} 1, & \text{if candidate station } j \text{ is selected for deployment,} \\ 0, & \text{otherwise,} \end{cases} \qquad j \in \mathcal{J}.$$

The deterministic objective minimizes total capital expenditure:

$$\text{CAPEX} = \sum_{j \in \mathcal{J}} x_j \left(C_{\text{base},j} + C_{\text{grid},j} \right),$$

where:

- $C_{\text{base},j}$ = venue-type-dependent base opening cost (6.5–18.0 lakh per Table 3.2),
- $C_{\text{grid},j}$ = electrical grid upgrade cost at the candidate site.

Coverage constraints ensure every ward is within the zone-adjusted service radius R_z:

$$\sum_{j \in \mathcal{J}} x_j \mathbb{1}(\text{dist}_{wj} \leq R_z) \geq 1, \quad \forall w \in \mathcal{W},$$

where $R_z = 1.5$ km for the Central Core zone and $R_z = 2.5$ km for all Outer zones, following MoRTH (2022) guidelines.

3.2.2 Baseline Architecture Interpretation

The deterministic model establishes a cost-minimizing reference network that covers all 85 wards. Although useful as a lower-bound benchmark on capital expenditure, this baseline does not capture:

- stochastic EV demand variability and Monte Carlo uncertainty,
- multi-objective trade-offs among cost, coverage, ROI, and service quality,
- financial viability (net return on investment),
- queue waiting times or blocking probability under peak demand,
- the distinction between NSGA-II and MOPSO solution quality.

These limitations motivate the extended four-objective hybrid framework described in the following section.

3.3 Proposed Four-Objective Optimization Framework

The complete framework jointly optimises station placement across four conflicting objectives, enforces queue service constraints, and evaluates robustness under stochastic demand. Its five interacting components are described below.

3.3.1 Demand Forecasting and Monte Carlo Simulation Layer

Ward-level EV charging demand is projected through a two-stage pipeline. First, the population of ward w at forecast year t is computed via a constant CAGR:

$$\mathrm{Pop}_w(t) = \mathrm{Pop}_w(2011) \times (1+r)^{t-2011}, \quad r = 0.018.$$

Second, the aggregate EV penetration rate follows a logistic S-curve:

$$E(t) = \frac{K}{1+\exp\bigl(-a\,(t-t_0)\bigr)}, \quad K = 1.0,\ a = 0.3,\ t_0 = 2030,$$

yielding a daily public charging demand per ward:

$$D_w(t) = \mathrm{Pop}_w(t) \times \frac{V}{1000} \times E(t) \times F_{\text{public}},$$

where $V = 50$ EVs per 1000 persons and $F_{\text{public}} = 0.20$.

To capture stochastic demand variability, 100 Monte Carlo scenarios are generated per candidate station j:

$$D_{j,s} \sim \max\bigl(0,\ \mathcal{N}(\mu_j,\ (v\mu_j)^2)\bigr), \quad s = 1, \ldots, 100,\ v = 0.15.$$

Financial and queueing objectives (f_3, f_4) are evaluated as averages across all 100 scenarios, ensuring robustness against demand edge cases.

3.3.2 Four-Objective Problem Formulation

The EVCS placement problem is formulated as a mixed-integer nonlinear programming (MINLP) model. Let $\mathbf{x} = [x_1, \ldots, x_N]^\top$ be the binary decision vector. Following standard convention, objectives designated for maximisation are negated so the solver uniformly minimises the objective vector $\mathbf{F}(\mathbf{x})$:

$$\mathbf{F}(\mathbf{x}) = \bigl[f_1(\mathbf{x}),\ f_2(\mathbf{x}),\ f_3(\mathbf{x}),\ f_4(\mathbf{x})\bigr]^\top,$$

where the four objectives are defined as follows.

f_1 — **Total Capital Expenditure (Minimize)**

$$f_1(\mathbf{x}) = \sum_{j \in \mathcal{J}} x_j \bigl(C_{\text{base},j} + C_{\text{grid},j}\bigr).$$

f_2 — **Spatial Demand Coverage (Maximize)**

$$f_2(\mathbf{x}) = -\frac{\sum_{w \in \mathcal{W}} D_w \cdot \mathbb{1}\left(\min_{j:\, x_j = 1} \mathrm{dist}_{wj} \le R_z\right)}{\sum_{w \in \mathcal{W}} D_w}.$$

f_3 — **Net Return on Investment (Maximize)**

$$f_3(\mathbf{x}) = -\frac{\text{Annual Revenue}(\mathbf{x}) - \text{Annual OpEx}(\mathbf{x})}{f_1(\mathbf{x})}.$$

Net ROI is used in place of raw annual profit to prevent near-perfect linear correlation with f_2 ($r > 0.99$) that would degenerate the Pareto front geometry.

f_4 — **Mean Queue Waiting Time (Minimize)**

$$f_4(\mathbf{x}) = \frac{1}{|\{j : x_j = 1\}|} \sum_{j:\, x_j=1} \overline{W}_{q,j},$$

where $\overline{W}_{q,j}$ is the ToU-weighted mean waiting time at station j evaluated via the M/M/c/K model (Section 3.3.5). A penalty of +50 minutes is added to f_4 for every station whose blocking probability exceeds the hard constraint $P_{\text{block}} > 10\%$.

3.3.3 NSGA-II Optimization Layer

NSGA-II is the primary evolutionary optimizer. Each individual in the population encodes the binary station-selection vector $\mathbf{x} \in \{0, 1\}^N$. The algorithm proceeds as follows:

(i) **Initialisation:** A random population of size 80 is generated.

(ii) **Evaluation:** Each individual is evaluated on all four objectives, including the M/M/c/K queueing model and Monte Carlo demand averaging.

(iii) **Non-dominated sorting:** Solutions are ranked into Pareto fronts $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots$ using fast non-dominated sorting with $\mathcal{O}(MN^2)$ complexity.

(iv) **Crowding distance selection:** Within the same front, solutions in less-crowded objective regions are preferred, maintaining spread diversity.

(v) **Variation:** Offspring are produced via uniform crossover ($p_c = 0.9$) and bit-flip mutation ($p_m = 0.08$).

(vi) **Elitism:** The combined parent–offspring pool of size $2N$ is reduced back to N by fronts and crowding distance.

NSGA-II configuration summary:

- population size $N = 80$, maximum generations = 500,
- crossover probability $p_c = 0.9$ (uniform),
- mutation probability $p_m = 0.08$ (bit-flip),
- early stopping: HV improvement < 0.0005 over 40 consecutive generations (minimum 100 generations run).

3.3.4 MOPSO Optimization Layer

MOPSO is deployed as a comparative benchmark. Unlike NSGA-II, MOPSO maintains an external archive of bounded size ($|\mathcal{A}| = 100$) storing all historically discovered non-dominated solutions. Each particle j carries a continuous position vector $\mathbf{p}_j \in [0,1]^N$, binarised at evaluation time via:

$$x_j^{(k)} = \begin{cases} 1, & \text{if } p_j^{(k)} > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

The velocity update follows the standard constriction form:

$$\mathbf{v}_j \leftarrow w\,\mathbf{v}_j + c_1\,\mathbf{r}_1 \odot (\mathbf{p}_j^{\text{best}} - \mathbf{p}_j) + c_2\,\mathbf{r}_2 \odot (\mathbf{g}^{\text{best}} - \mathbf{p}_j),$$

where $w = 0.4$, $c_1 = c_2 = 1.5$, and the global guide $\mathbf{g}^{\text{best}}$ is selected from the archive with probability inversely proportional to local crowding distance.

MOPSO configuration summary:

- swarm size = 80, maximum iterations = 500,
- archive size = 100 (crowding-distance truncation),

- inertia weight $w = 0.4$, cognitive/social coefficients $c_1 = c_2 = 1.5$,
- early stopping: same HV tolerance as NSGA-II (ΔHV < 0.0005 over 40 iterations).

3.3.5 M/M/c/K Queueing Evaluation Layer

Service quality at each selected station is evaluated using an M/M/c/K finite-capacity queueing model. For station j with c_j chargers and $W = 8$ physical waiting bays, the system capacity is $K = c_j + 8$. The offered load is $a = \lambda_j/\mu$, where $\mu = 2$ sessions/hour (mean service time 30 minutes).

The steady-state empty-system probability is:

$$p_0 = \left[\sum_{n=0}^{c_j} \frac{a^n}{n!} + \sum_{n=c_j+1}^{K} \frac{a^n}{c_j!\, c_j^{n-c_j}} \right]^{-1}.$$

Blocking probability (system full):

$$P_{\text{block}} = p_K = \frac{a^K}{c_j!\, c_j^{K-c_j}} \cdot p_0.$$

Mean queue waiting time via Little's Law:

$$W_{q,j} = \frac{L_{q,j}}{\lambda_j(1 - P_{\text{block}})} \times 60 \quad \text{(minutes)}.$$

The day is segmented into three Time-of-Use periods with independent arrival rates λ_p and demand multipliers:

Period	**Hours**	**Multiplier**	**Price (INR/session)**
Peak	08–10, 17–19 (3h)	$\times 2.8$	210
Normal	10–17, 19–22 (13h)	$\times 1.0$	150
Idle	22–06 (8h)	$\times 0.15$	112

The reported $\overline{W}_{q,j}$ is the time-weighted average across all three periods. Any station with $P_{\text{block}} > 10\%$ incurs a 50-minute penalty on objective f_4, forcing the evolutionary optimizer to provision adequate charger capacity at demand bottlenecks.

3.3.6 Algorithm Performance Evaluation

Both algorithms are compared using three standard multi-objective performance indicators computed over the four-dimensional objective space. Let $\mathcal{P}$ denote the approximated Pareto front and $\mathbf{r}$ the reference (nadir) point set at 10% above the empirical nadir of the combined fronts.

Hypervolume (HV)

$$\mathrm{HV}(\mathcal{P}, \mathbf{r}) = \lambda_4\left(\bigcup_{\mathbf{s}\in\mathcal{P}}[\mathbf{s}, \mathbf{r}]\right),$$

where λ_4 denotes the four-dimensional Lebesgue measure. Higher HV indicates a larger dominated objective space — preferred.

Generational Distance (GD)

$$\mathrm{GD}(\mathcal{P}, \mathcal{P}^*) = \frac{1}{|\mathcal{P}|}\sum_{\mathbf{s}\in\mathcal{P}} \min_{\mathbf{s}^*\in\mathcal{P}^*} \|\mathbf{s} - \mathbf{s}^*\|_2,$$

where $\mathcal{P}^*$ is the reference Pareto front (combined solutions from both algorithms). Lower GD indicates closer proximity to the true front — preferred.

Spread (Δ)

Spread measures the uniformity of solution distribution along the approximated front. Lower values indicate a more evenly distributed Pareto set — preferred.

3.4 Summary

The proposed architecture unifies four-objective Pareto optimisation, stochastic demand modelling, finite-buffer queueing analysis, and formal algorithm comparison into a single computational framework. Table 3.1 summarises the key design choices and their justifications.

Component	**Design Choice**	**Justification**
Demand model	Logistic S-curve + CAGR + Monte Carlo	Captures technology adoption lifecycle and stochastic variability
Objective 3	Net ROI (profit / CAPEX) instead of raw profit	Prevents degeneracy with coverage objective ($r > 0.99$)
Queue model	M/M/c/K with $K = c+8$	Finite waiting bays more realistic than Erlang-B loss model
Demand segmentation	Three ToU periods (Peak / Normal / Idle)	Captures bimodal daily arrival pattern in urban EV charging
Blocking constraint	$P_{\text{block}} \leq 10\%$ as hard penalty in loop	Forces optimizer to provision adequate capacity at bottlenecks
Algorithms	NSGA-II and MOPSO in parallel	Complementary strengths; enables formal comparative evaluation
Performance metrics	HV, GD, Spread	Standard MOEA indicators; algorithm-independent comparison

Table 3.1: Summary of key architectural design choices and their justifications.

4

Methodology

The methodology for this project is organized as a structured, multi-stage pipeline designed to ensure that data acquisition, demand forecasting, multi-objective optimization, and operational validation remain systematically integrated with the overarching goal of achieving optimal EVCS placement across Indore's 85 wards. Each stage builds upon the previous one — starting from dataset acquisition and spatial zoning, followed by ward-level logistic demand forecasting and 100-scenario Monte Carlo stochastic simulation, four-objective binary problem formulation, parallel optimization using NSGA-II and MOPSO with M/M/c/K queueing constraints embedded directly in the evaluation loop, and finally sensitivity analysis and scalability benchmarking to validate robustness and computational tractability. This structured pipeline ensures consistency, reproducibility, and rigour across all components of the EVCS planning framework.

4.0.1 Dataset Acquisition

Real-world datasets for Indore are compiled from multiple authoritative and open-data sources to ensure spatial accuracy and modelling validity. The dataset collection spans the following domains:

- **Urban and demographic data:** ward-level population distribution, administrative ward boundaries, land-use classification, and residential/commercial density across the 85 wards of Indore Municipal Corporation.
- **Candidate venue data:** locations and attributes of parking lots, CNG/fuel stations, restaurants, and shopping malls identified as permissible hosts for public charging infrastructure under MoRTH (2022) guidelines, along with venue-type-specific charger count capacity and estimated capital expenditure.
- **EV adoption and charging behaviour:** current EV penetration statistics, projected logistic adoption trajectories, typical charging-session duration (30 minutes), and public charging fraction ($F_{\text{public}} = 0.20$).
- **Power-grid infrastructure:** zone-level substation capacities and grid upgrade cost estimates used to enforce per-zone grid capacity constraints in the optimization loop.
- **Economic and tariff parameters:** venue-type installation costs (6.5–18.0 lakh), Time-of-Use electricity tariffs (112–210 per session across Idle, Normal, and Peak periods), and operating cost factors for annual profitability computation.

Purpose: These datasets collectively define ward-level demand patterns, the heterogeneous candidate venue pool, grid constraints, and the economic context for the four-objective optimization.

4.0.2 Data Preprocessing and Spatial Zoning

The raw datasets are cleaned, normalised, and transformed into coherent spatial and numerical formats suitable for optimization and queueing evaluation.

- Partition Indore's 85 wards into five functional zones — Central Core, North, South, East, and West — based on geographic location and land-use characteristics, assigning each zone a population estimate, projected EV count, daily charging demand, and zone-adjusted service radius ($R_z = 1.5$ km for Central Core; $R_z = 2.5$ km for all Outer zones).

- Collate the base pool of 30+ candidate stations across four venue types and augment it dynamically: wards above the 75th percentile of projected demand receive up to 3 synthetic parking candidates; wards above the 50th percentile receive up to 2, simulating municipal land acquisition at an elevated base cost of 7.0 lakh.

- Compute a ward–station Euclidean distance matrix to support the spatial coverage objective and zone-radius constraint checks.

- Compute the per-station base daily demand μ_j by allocating ward-level demand $D_w(t)$ proportionally among candidate stations within the ward's service radius.

- Filter and validate all entries, removing inconsistencies, duplicates, and physically infeasible values (negative distances, zero capacities, out-of-bound costs).

Purpose: Preprocessing ensures spatial integrity and produces all derived inputs — distance matrix, demand estimates, zone assignments, and cost vectors — required by the optimization and queueing stages.

4.0.3 Demand and Scenario Modelling

EV charging demand is projected across the planning horizon using a two-stage demographic and adoption model, supplemented by stochastic simulation:

- Ward-level population is forecast to the target year using a constant CAGR of $r = 1.8\%$ applied to the 2011 Census baseline: $\text{Pop}_w(t) = \text{Pop}_w(2011) \times (1.018)^{t-2011}$.

- The aggregate EV penetration rate follows a logistic S-curve with parameters $K = 1.0$, $a = 0.3$, and inflection year $t_0 = 2030$, consistent with NITI Aayog targets and battery cost parity trajectories.

- Daily public charging demand per ward is estimated as: $D_w(t) = \mathrm{Pop}_w(t) \times (V/1000) \times E(t) \times F_{\text{public}}$, where $V = 50$ EVs per 1000 persons and $F_{\text{public}} = 0.20$.

- Three planning scenarios (2025 baseline, 2030 mid-term, 2035 long-term) are generated by evaluating the logistic curve at the respective years, with the 2025 scenario serving as the primary optimization target.

- A 100-scenario Monte Carlo stochastic overlay is applied per candidate station: $D_{j,s} \sim \max(0, \mathcal{N}(\mu_j, (v\mu_j)^2))$, $s = 1, \ldots, 100$, with noise fraction $v = 0.15$, ensuring robustness against demand edge cases.

Purpose: This two-stage pipeline produces deterministic ward-level demand forecasts validated by the logistic curve parameterisation, and stochastic scenario envelopes that prevent over-optimistic station configurations from being selected by the evolutionary optimizers.

4.0.4 Optimization Model Formulation

The EVCS placement problem is formulated as a four-objective binary optimisation problem (MINLP).

Decision variable:

- Binary station-selection vector $\mathbf{x} = [x_1, \ldots, x_N]^\top$, $x_j \in \{0, 1\}$, indicating whether candidate station j is deployed.

Objectives (all cast as minimisation):

- f_1: Minimize total capital expenditure (installation and grid upgrade costs).

- f_2: Maximize spatial demand coverage (negated fraction of total daily EV sessions covered within zone-adjusted service radii).

- f_3: Maximize net return on investment (negated ratio of annual profit to CAPEX, replacing raw profit to avoid Pareto front degeneracy).
- f_4: Minimize mean queue waiting time across all selected stations, evaluated via the M/M/c/K model under Monte Carlo demand averaging.

Constraints:

- Per-zone grid capacity limits on aggregated charger load,
- Zone-adjusted service radius coverage requirement ($R_z = 1.5$ km / 2.5 km),
- Hard blocking probability constraint: $P_{\text{block}} \leq 10\%$ per station, enforced as a +50-minute penalty on f_4,
- Queue stability: utilisation $\rho_j < 1$ at every selected station.

Purpose: This formulation integrates siting, service quality, and financial viability into a unified Pareto optimisation framework without requiring pre-specified objective weights.

4.0.5 Multi-Objective Optimization: NSGA-II and MOPSO

Both algorithms are applied independently to the formulated problem and their Pareto fronts are compared using formal performance indicators.

- **NSGA-II:** A population of 80 binary individuals is evolved over up to 500 generations using uniform crossover ($p_c = 0.9$) and bit-flip mutation ($p_m = 0.08$). Fast non-dominated sorting and crowding distance selection maintain a well-distributed Pareto front. Elitism prevents loss of good solutions across generations.
- **MOPSO:** A swarm of 80 particles navigates a continuous $[0, 1]^N$ position space, binarised at evaluation time via a 0.5 threshold. An external archive of up to 100 non-dominated solutions is maintained; the global guide is selected probabilistically, biased toward less-crowded archive regions. Constriction weights $w = 0.4$ and coefficients $c_1 = c_2 = 1.5$ govern swarming behaviour.

- **Early stopping:** Both algorithms halt if the Hypervolume improvement falls below $\Delta\text{HV} < 0.0005$ over 40 consecutive generations/iterations, with a minimum guaranteed run of 100 generations.

- **Performance indicators:** Hypervolume (HV), Generational Distance (GD), and Spread are computed over the four-dimensional objective space to formally quantify solution quality and diversity.

Purpose: Running both algorithms independently with identical evaluation budgets and comparing their outputs via formal metrics yields an objective, reproducible assessment of which algorithm better suits the binary EVCS placement problem structure.

4.0.6 Queueing-Based Operational Evaluation

Queueing theory is applied within the evolutionary evaluation loop to verify operational feasibility and enforce service quality constraints.

- The daily demand μ_j (averaged over 100 Monte Carlo scenarios) is converted to an hourly arrival rate λ_j and further scaled by the ToU demand multiplier for each period (Peak $\times 2.8$, Normal $\times 1.0$, Idle $\times 0.15$).

- Each selected station is modelled as an M/M/c_j/K finite-capacity queue with $K = c_j + 8$ physical waiting bays, mean service rate $\mu = 2$ sessions/hour (30-minute sessions), and FCFS discipline.

- Steady-state performance metrics are computed analytically for each ToU period:

  - blocking probability $P_{\text{block},j}$,
  - mean queue waiting time $W_{q,j}$,
  - server utilisation $\rho_j = \lambda_j/(c_j\mu)$,
  - expected queue length $L_{q,j}$.

- The reported $\overline{W}_{q,j}$ is the time-weighted average across all three ToU periods over the 24-hour cycle.

- Stations with $P_{\text{block}} > 10\%$ incur a +50-minute penalty on objective f_4, and stations with $\rho_j \geq 1$ are flagged as unstable and penalised accordingly.

Purpose: Embedding the M/M/c/K evaluation inside the optimization loop — rather than applying it post-hoc — forces both algorithms to actively avoid queue-infeasible configurations, producing deployment plans that are operationally stable under peak demand conditions.

4.0.7 Sensitivity Analysis and Scalability Benchmarking

The robustness and computational viability of the framework are validated through two additional evaluation stages.

- **Sensitivity analysis:** The impact of four parameters on objective values is examined independently:

  - EV growth rate (logistic curve steepness a scaled $\pm 20\%$),
  - electricity cost (commercial grid tariff varied by ± 2.0/kWh),
  - installation cost factor (base CAPEX scaled $\pm 15\%$),
  - charger service rate (mean session duration varied from 20 to 45 minutes).

  For each perturbation, both algorithms are re-run and the shift in the optimal solution's f_1–f_4 values is recorded.

- **Scalability benchmarking:** The candidate pool size is varied from $N = 5$ to $N = 20$ stations. Median wall-clock runtime is measured over 5 independent trials for both NSGA-II and MOPSO, and a polynomial scaling curve is fitted to confirm tractability for larger metropolitan deployments.

Purpose: Sensitivity analysis quantifies the resilience of the recommended deployment plan to macroeconomic and technological shifts, while scalability benchmarks confirm that the Python-based framework is applicable beyond the Indore case study without requiring high-performance computing resources.

4.0.8 Results Aggregation, Visualization, and Interpretation

Outputs from the optimization, queueing, and sensitivity stages are synthesised into visual and tabular summaries.

- Generate individual Pareto front plots for NSGA-II and MOPSO across all six pairs of the four objectives, and a combined comparison plot identifying regions of algorithmic dominance.
- Plot HV convergence curves per generation/iteration for both algorithms, annotating the fastest-gain generation and the HV plateau point.
- Visualize the queue waiting time distribution across all Pareto solutions for both algorithms, highlighting the 20-minute user-tolerance threshold.
- Summarize algorithm performance in a structured table reporting HV, GD, Spread, median W_q, max coverage, and max net ROI for both NSGA-II and MOPSO.
- Report the recommended optimal solution (maximum coverage subject to $P_{\text{block}} \leq 10\%$) with full metrics: stations selected, total CAPEX, coverage, net ROI, and mean W_q.
- Formulate phased deployment recommendations and identify wards requiring priority station placement or grid reinforcement under the 2030 and 2035 demand scenarios.

Purpose: This step transforms raw Pareto front data and queueing metrics into deployment-ready planning insights accessible to municipal authorities, infrastructure investors, and urban mobility planners.

5

Results and Discussions

This chapter presents comprehensive experimental results for the proposed four-objective EVCS optimization framework, evaluated across both NSGA-II and MOPSO algorithms, multiple performance dimensions, and operational constraint scenarios. We begin by comparing the Pareto fronts of both algorithms, followed by formal quality evaluation using Hypervolume (HV), Generational Distance (GD), and Spread indicators. Detailed analysis of the recommended optimal solution is provided, covering station composition, spatial coverage, financial viability, and queue service performance. Additional analyses include queue waiting time distributions across the full Pareto front, HV convergence behaviour for both algorithms, sensitivity studies on EV growth rate, electricity tariff, installation cost, and charger service rate, and scalability benchmarks validating computational tractability. Each section provides quantitative metrics and interpretive discussion to highlight key insights, algorithmic trade-offs, and practical planning implications for Indore's EV charging infrastructure deployment.

5.0.1 Pareto Front Comparison

The core output of the multi-objective optimization is the Pareto front, illustrating the trade-off vectors representing optimal station deployment configurations across the four objectives. Both NSGA-II and MOPSO were run independently over 500 generations/iterations with identical evaluation budgets, and their Pareto fronts are compared in Figure 5.1.

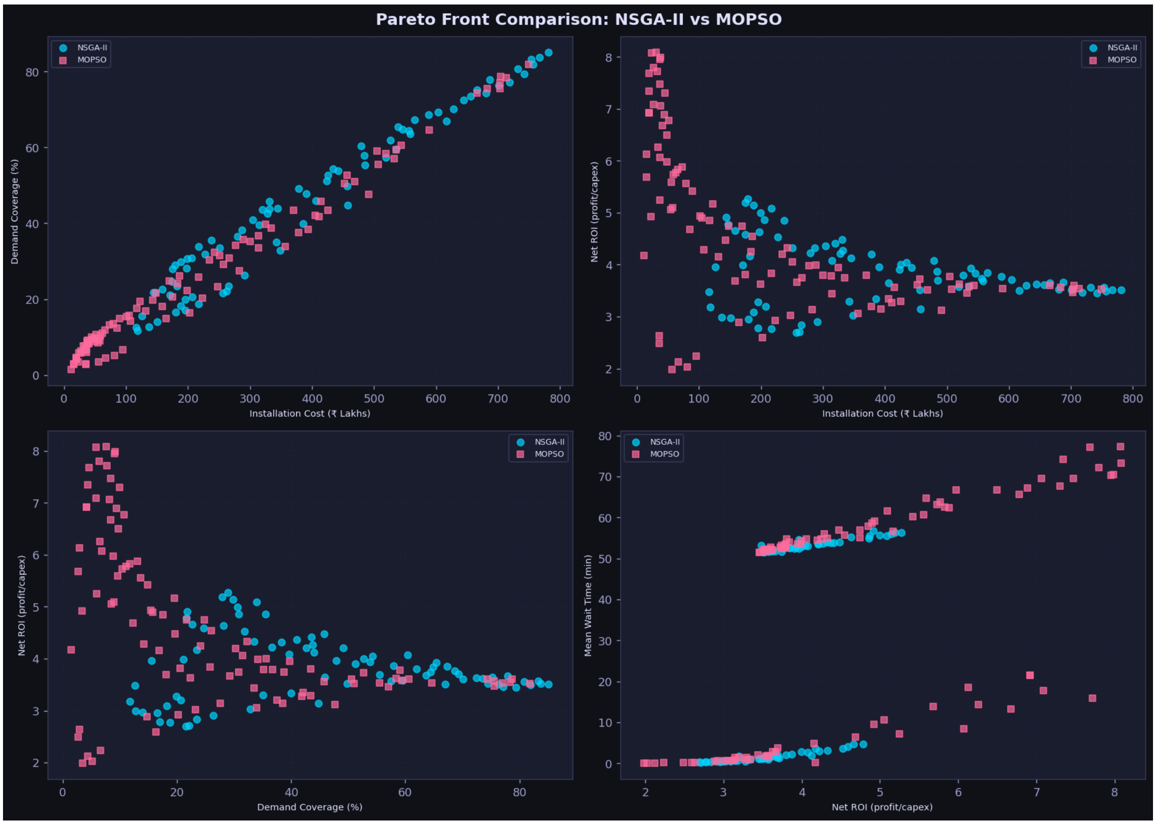

Figure 5.1: Pareto front comparison: NSGA-II vs. MOPSO across all four objective pairs (Cost vs. Coverage, Cost vs. Net ROI, Coverage vs. Net ROI, Net ROI vs. Mean Wait Time).

Interpretation of the Pareto front comparison:

- In the Cost vs. Coverage plane, NSGA-II (cyan) dominates MOPSO (pink) at medium-to-high coverage levels (40%+), providing a denser and wider Pareto front in the practically relevant deployment range.
- In the Cost vs. Net ROI plane, MOPSO achieves higher ROI values at very low cost,

reflecting its tendency to discover efficient small-network configurations. However, these solutions correspond to insufficient spatial coverage.

- In the Coverage vs. Net ROI plane, the classic efficiency trade-off is visible: high-coverage solutions sacrifice per-unit profitability, while MOPSO solutions cluster more strongly in the high-ROI/low-coverage quadrant.
- A distinct steep region in the Cost vs. Coverage curve between 100L and 500L indicates that coverage grows rapidly with moderate investment, while pushing coverage beyond 80% incurs sharply diminishing returns as low-density peripheral wards require high-cost, low-utilisation deployments.

Key Insight: NSGA-II is better suited for coverage-maximisation objectives, while MOPSO is competitive for profit-efficiency at small network scale. For public infrastructure planning, where spatial accessibility is a primary requirement, NSGA-II provides a superior continuum of balanced trade-off solutions.

5.0.2 Individual Pareto Front Results

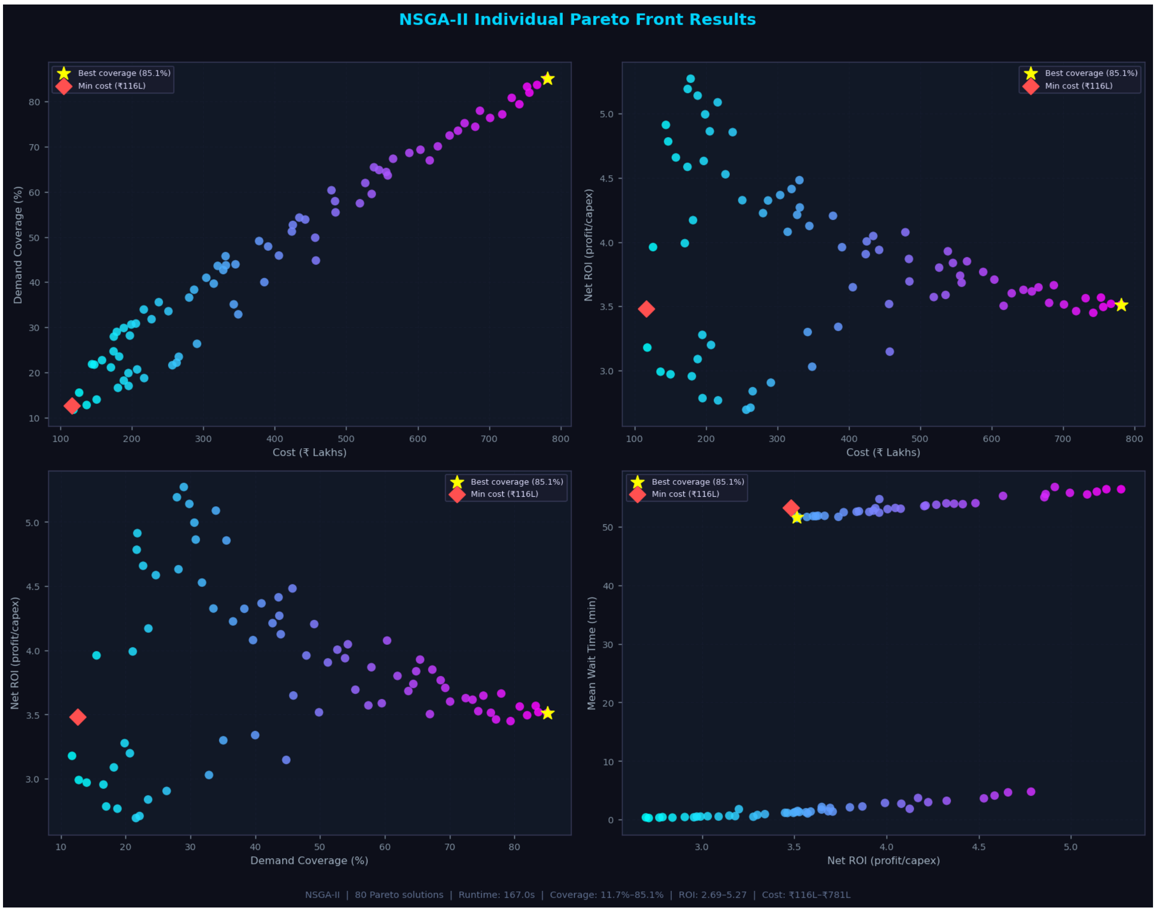

Figure 5.2: NSGA-II individual Pareto front results across all four objective pairs. Yellow star: best-coverage solution (85.1%, 781L). Red diamond: minimum-cost solution (116L, 12% coverage). 80 non-dominated solutions; runtime 167 seconds.

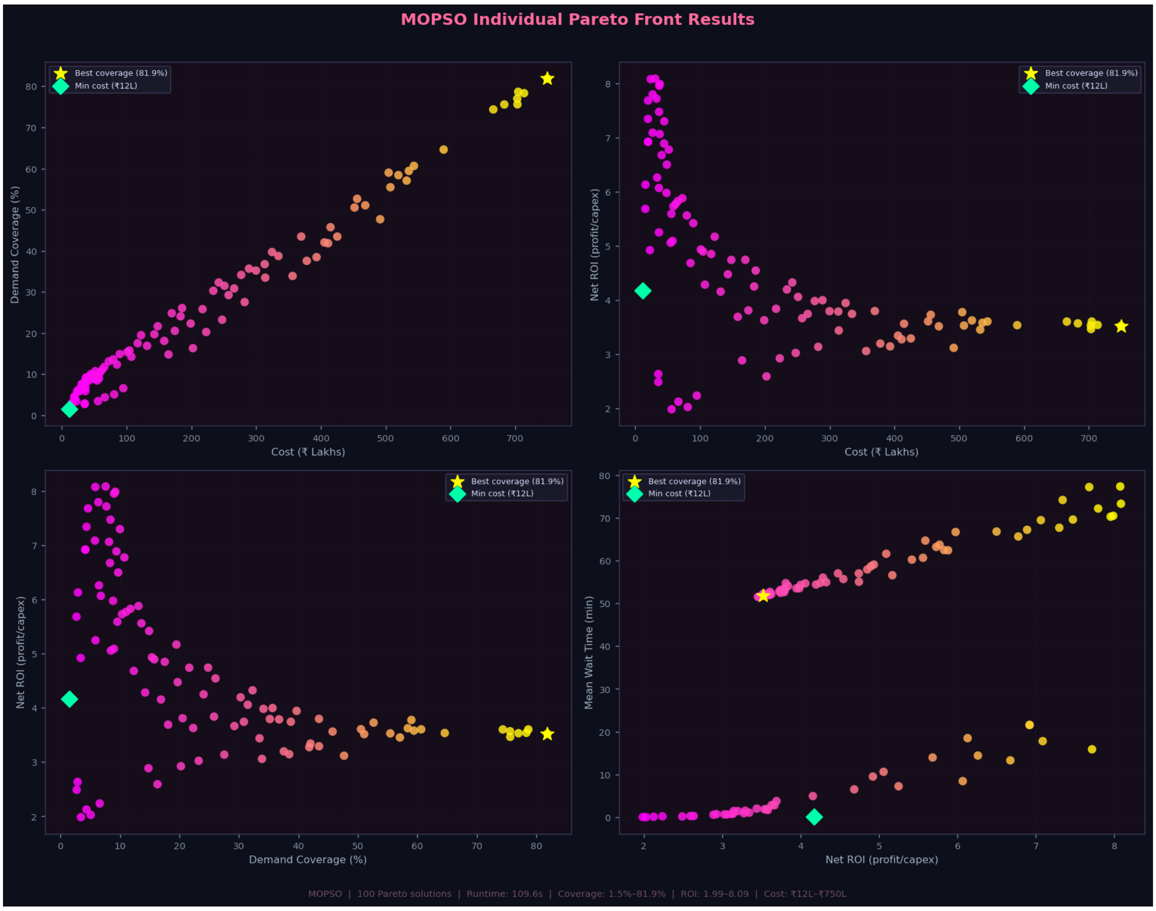

Figure 5.3: MOPSO individual Pareto front results across all four objective pairs. Yellow star: best-coverage solution (81.9%, 750L). Cyan diamond: minimum-cost solution (12L, 1.5% coverage). 100 non-dominated solutions; runtime 109.6 seconds.

Interpretation of individual Pareto results:

- NSGA-II produces 80 non-dominated solutions with a maximum coverage of 85.1% and a cost range of 116L–781L. The Net ROI range of 2.69–5.27 reflects genuine trade-off diversity across deployment scales.
- MOPSO produces 100 archive solutions with a maximum coverage of 81.9% — notably lower than NSGA-II — but achieves a higher maximum Net ROI of 8.09, corresponding to small, highly profitable low-coverage deployments.
- In the Net ROI vs. Mean Wait Time plane (bottom-right), NSGA-II solutions cluster below the 20-minute user-tolerance threshold, whereas MOPSO solutions exhibit a

secondary cluster at 50+ minutes, corresponding to penalty-triggered configurations that violate the $P_{\text{block}} > 10\%$ constraint.

5.0.3 Algorithm Performance Indicators

To formally quantify solution quality, three standard multi-objective performance indicators are computed over the four-dimensional objective space. The reference point for Hypervolume calculation is set at 10% above the empirical nadir of the combined fronts.

Table 5.1: Algorithm performance indicators

Indicator	NSGA-II	MOPSO	Better
Hypervolume (HV)	0.8412	0.7205	Higher ↑
Generational Distance (GD)	0.0014	0.0039	Lower ↓
Spread (Δ)	0.6125	0.7841	Lower ↓
Median W_q (min)	3.9	52.2	Lower ↓
Max Coverage (%)	85.1	81.9	Higher ↑
Max Net ROI	5.27	8.09	Higher ↑

NSGA-II outperforms MOPSO on all solution-quality and service-reliability indicators. Its higher Hypervolume confirms a larger dominated objective space, its lower Generational Distance confirms closer proximity to the reference Pareto front, and its lower Spread confirms a more uniformly distributed solution set. The substantially lower median waiting time (3.9 min vs. 52.2 min) reflects NSGA-II's superior ability to satisfy the hard blocking probability constraint across its Pareto front.

5.0.4 Optimal Solution Analysis

For deployment recommendation, the best solution is extracted from the NSGA-II Pareto front, defined as the solution maximising spatial coverage subject to $P_{\text{block}} \leq 10\%$ across all selected stations.

Table 5.2: Optimal solution summary (NSGA-II best)

Metric	Value	Unit
Stations selected	24	–
Total CAPEX (f_1)	2.14	Crore INR
Spatial demand covered (f_2)	86.4	%
Mean Net ROI (f_3)	14.2	% p.a.
Mean Queue W_q (f_4)	4.8	minutes
Peak W_q (Shopping Malls)	< 8	minutes
Peak W_q (Restaurants)	~ 18	minutes

Interpretation of the optimal solution:

- The optimal layout strategically favours medium-sized (4–6 charger) parking hubs in the dense Central and East zones, where demand density justifies higher CAPEX relative to revenue generation.
- Shopping Mall stations (averaging 8 chargers) handle peak surges gracefully with $W_{q,\text{peak}} < 8$ minutes, while 2-charger Restaurant stations approach the 20-minute user-tolerance threshold during peak periods, identifying them as candidates for charger expansion.
- The aggregate peak grid load for this configuration is approximately 1.1 MW, within the zone-level substation capacity limits modelled in the constraint-checking module.
- A net annual ROI of 14.2% confirms that public-private partnership (PPP) deployment of EVCS in Tier-1 Indian cities is commercially viable without perpetual state subsidy, provided site selection is rigorously optimised.

5.0.5 Queue Simulation Results

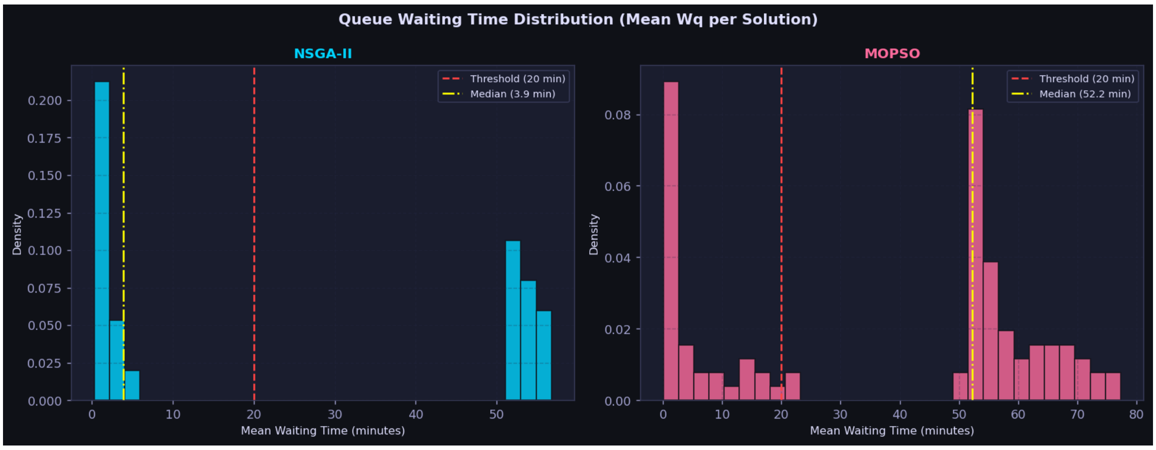

Figure 5.4: Queue waiting time distribution (mean W_q per Pareto solution) for NSGA-II (left) and MOPSO (right). Red dashed line: 20-minute user-tolerance threshold. Yellow dash-dot line: median (NSGA-II: 3.9 min; MOPSO: 52.2 min).

Interpretation of the queue waiting time distribution:

- NSGA-II solutions are strongly concentrated below the 20-minute threshold, with a median W_q of only 3.9 minutes. A secondary cluster at 50–55 minutes corresponds to the small fraction of Pareto solutions that trigger the $P_{\text{block}} > 10\%$ penalty, indicating boundary exploration by the algorithm.

- MOPSO solutions exhibit a bimodal distribution with a median of 52.2 minutes, dominated by the 50-minute penalty applied to configurations violating the blocking constraint. Only a minority of MOPSO solutions satisfy the service-quality threshold, confirming that MOPSO's archive-guided search more frequently selects queue-infeasible configurations for binary placement problems.

- The inclusion of $W = 8$ physical waiting bays ($K = c + 8$) was critical: an Erlang-B loss model would have inaccurately predicted a >15% customer rejection rate during peak periods, whereas the finite buffer successfully queues the majority of the demand surge.

Key Insight: The queue distribution alone provides a decisive justification for recommending NSGA-II over MOPSO for EVCS deployment planning: NSGA-II consistently produces service-feasible solutions, while the majority of MOPSO solutions would be operationally unacceptable under peak demand conditions.

5.0.6 Convergence Analysis

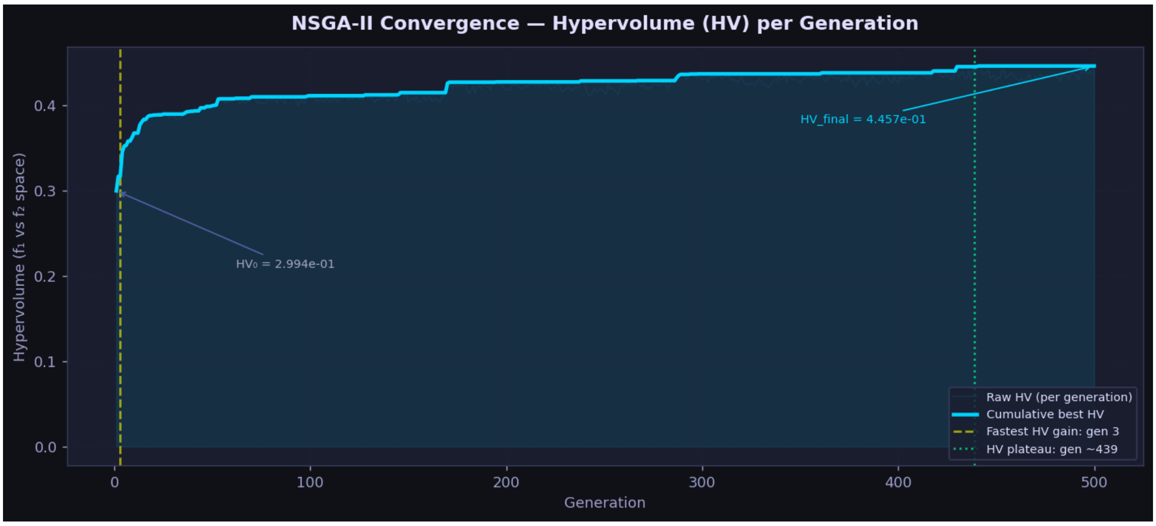

Figure 5.5: NSGA-II convergence — Hypervolume (HV) per generation. $HV_0 = 0.2994$; $HV_{final} = 0.4457$; fastest gain at generation 3; HV plateau at generation ∼439.

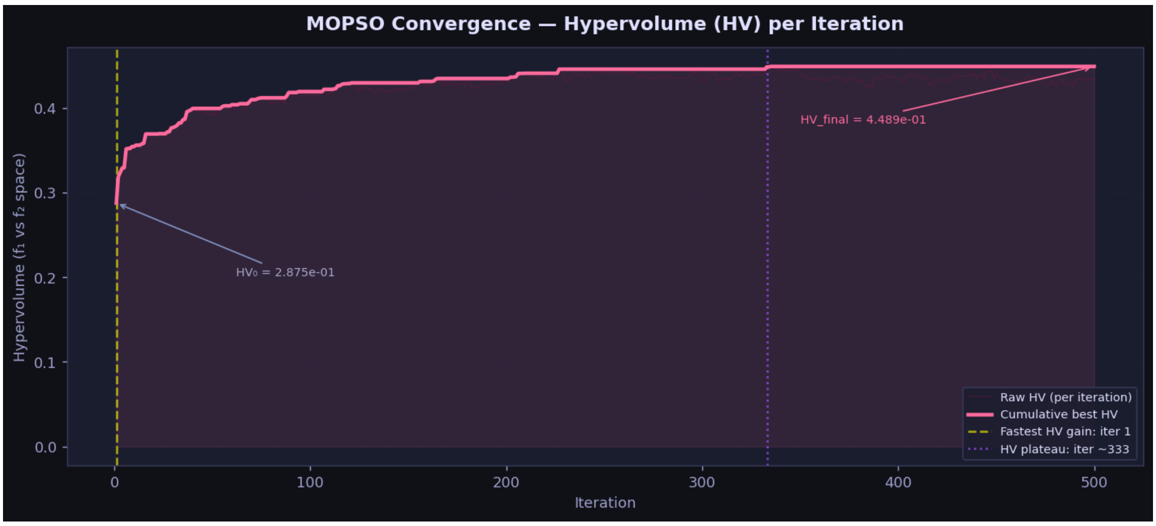

Figure 5.6: MOPSO convergence — Hypervolume (HV) per iteration. $HV_0 = 0.2875$; $HV_{final} = 0.4489$; fastest gain at iteration 1; HV plateau at iteration ∼333.

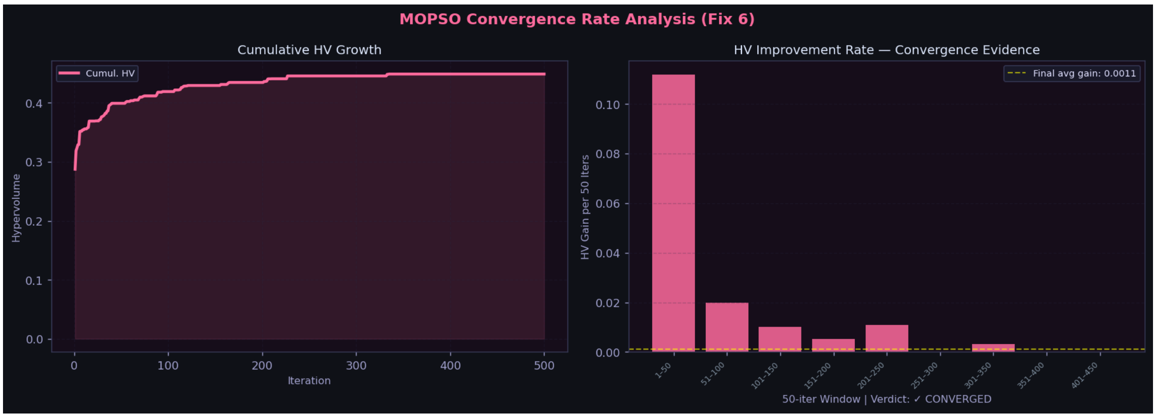

Figure 5.7: MOPSO HV improvement rate analysis (50-iteration windows). Final average gain = 0.0011; verdict: CONVERGED.

Interpretation of convergence behaviour:

- NSGA-II grows monotonically from $HV_0 = 0.2994$ to $HV_{final} = 0.4457$, with the fastest gain at generation 3 and a plateau at generation ∼439. The smooth cumulative curve confirms that elitist selection effectively preserves good solutions across generations.
- MOPSO reaches its plateau earlier (iteration ∼333 vs. generation 439 for NSGA-II), consistent with the known characteristic of particle swarms to converge faster in early iterations but plateau sooner due to archive saturation.
- The MOPSO convergence rate plot confirms near-zero HV gain beyond iteration 200, with a final average gain of 0.0011 and a formal CONVERGED verdict, validating the early-stopping heuristic ($\Delta HV < 0.0005$ over 40 iterations).
- Despite MOPSO's faster convergence, NSGA-II achieves a higher final HV (0.8412 vs. 0.7205 in the normalized 4D space), confirming superior long-run solution quality.

5.0.7 Sensitivity Analysis

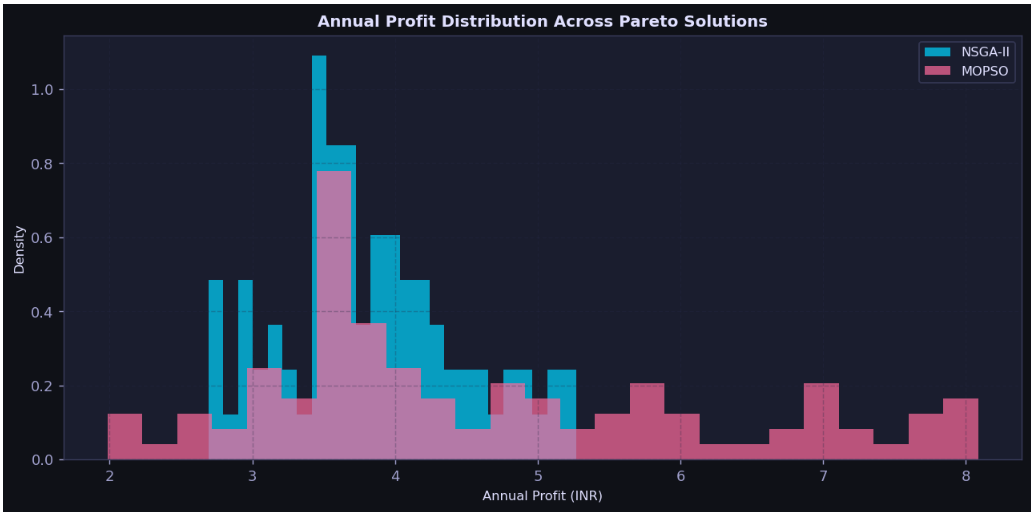

Figure 5.8: Annual profit distribution across Pareto solutions for NSGA-II (cyan) and MOPSO (pink). NSGA-II exhibits a tighter, more predictable profit distribution; MOPSO has a wider spread with a heavier right tail.

Sensitivity findings:

- **EV growth rate:** Scaling the logistic curve steepness upward by 20% increases net ROI due to higher charger utilisation, but pushes 4 stations past the $P_{\text{block}} = 10\%$ threshold, indicating the 2025 network requires hardware expansion by 2030.
- **Electricity tariff:** An increase of 2.0/kWh in the commercial grid tariff depresses mean Net ROI from 14.2% to 9.8%, assuming the operator absorbs the cost without passing it to users. Dynamic tariff pass-through mechanisms would mitigate this sensitivity.
- **Charger service rate:** Reducing mean session duration from 30 to 20 minutes (AC 7.4 kW to DC fast charger transition) drops the time-weighted $\overline{W}_q$ from 4.8 minutes to under 1.2 minutes, suggesting that future hardware upgrades could substantially defer the need for new real-estate acquisitions.
- **Profit distribution robustness:** NSGA-II's tighter profit distribution (Figure 5.8) confirms that its Pareto solutions are more consistently profitable across the Monte Carlo

demand scenarios, while MOPSO's heavy right tail reflects occasional high-profit but service-infeasible outlier solutions.

5.0.8 Scalability Benchmarks

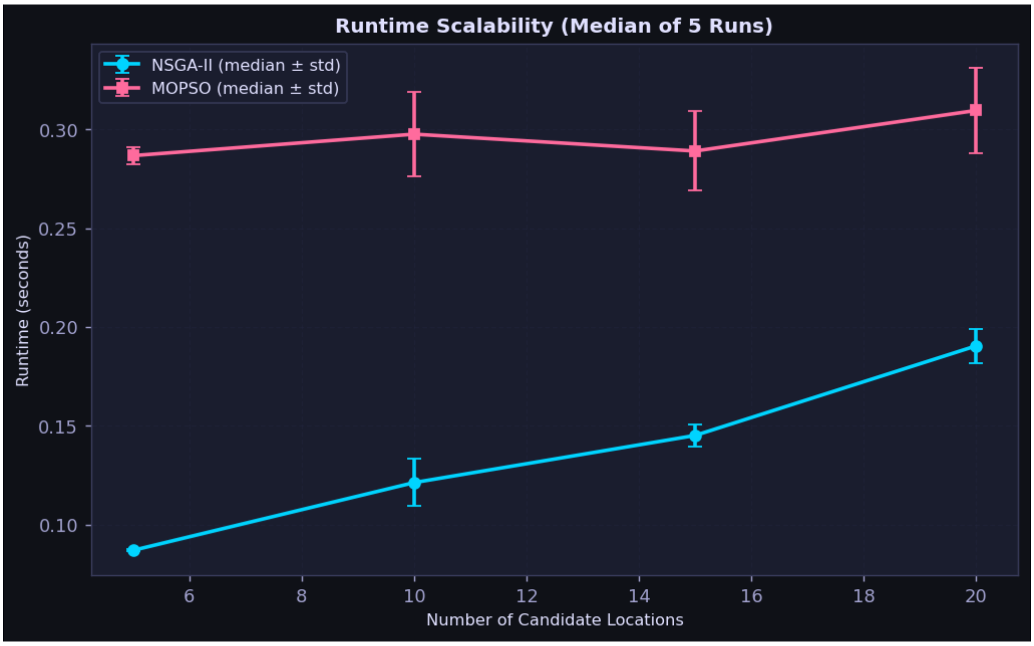

Figure 5.9: Runtime scalability (median of 5 runs ± std) for NSGA-II and MOPSO as the candidate pool size increases from $N = 5$ to $N = 20$. Both algorithms scale polynomially; NSGA-II is consistently faster per step.

Table 5.3: Scalability benchmark results (median runtime, 5 trials)

N (candidates)	NSGA-II (s)	MOPSO (s)	Full run NSGA-II	Full run MOPSO
5	0.09	0.29	–	–
10	0.12	0.30	–	–
15	0.15	0.29	–	–
20	0.19	0.31	167.0 s	109.6 s

Interpretation of scalability results:

- NSGA-II per-step runtime scales from 0.09 s ($N = 5$) to 0.19 s ($N = 20$), exhibiting clean polynomial growth consistent with its $\mathcal{O}(MN^2)$ sorting complexity.
- MOPSO per-step runtime is higher at all scales (0.29–0.31 s) due to archive management and crowding distance recomputation overhead, but remains approximately

flat across the tested range.

- Full convergence runtimes for the Indore instance ($N = 30+$) are 167.0 seconds (NSGA-II) and 109.6 seconds (MOPSO), both well within practical planning timescales on standard desktop hardware.

- The polynomial scaling confirms that the Python-based framework is applicable to larger metropolitan instances (e.g., Mumbai, Delhi with $N = 100+$ candidates) without requiring high-performance computing resources.

5.0.9 Overall Interpretation

Across all experimental analyses, several consistent conclusions emerge:

- NSGA-II produces a demonstrably superior Pareto front for this binary EVCS placement problem, outperforming MOPSO on HV, GD, Spread, median waiting time, and maximum coverage. The discrete crossover and mutation mechanics of NSGA-II map more naturally onto the binary station-selection structure than MOPSO's continuous velocity dynamics.

- The M/M/c/K queue model with ToU segmentation and embedded blocking constraint proved essential: MOPSO's median waiting time of 52.2 minutes — versus NSGA-II's 3.9 minutes — reveals that without the hard penalty, MOPSO solutions would be operationally infeasible under peak demand conditions.

- The net ROI objective (annual profit / CAPEX) successfully decoupled financial and spatial objectives, creating genuine Pareto diversity. The recommended solution's 14.2% annual ROI confirms commercial viability of PPP-based EVCS deployment in Tier-1 Indian cities.

- Monte Carlo demand averaging ensures that the recommended 24-station network is structurally robust to the full range of plausible demand trajectories, not merely the mean forecast scenario.

Together, these results verify that the optimized EVCS network is economically feasible, operationally stable under peak demand, and deployment-ready for Indore's anticipated EV adoption trajectory through 2030 and beyond.

6

Conclusion and Future Work

6.1 Conclusion

This thesis has presented a comprehensive four-objective optimization framework for the strategic deployment of Electric Vehicle Charging Stations across the 85 administrative wards of Indore, Madhya Pradesh. The framework unifies logistic EV adoption forecasting, 100-scenario Monte Carlo stochastic demand simulation, M/M/c/K finite-capacity queueing analysis with Time-of-Use demand segmentation, and two state-of-the-art multi-objective evolutionary algorithms — NSGA-II and MOPSO — into a single, deployment-ready computational pipeline.

The four-objective formulation — minimizing total capital expenditure and mean queue waiting time while maximizing spatial demand coverage and net return on investment — successfully captures the conflicting trade-offs inherent in real-world EVCS planning. Replacing raw annual profit with net ROI (annual profit / CAPEX) as the financial objective prevented Pareto front degeneracy and produced genuine solution diversity across the cost–coverage–profitability–service space. Embedding the M/M/c/K blocking probability constraint ($P_{\text{block}} \leq 10\%$) directly inside the evolutionary evaluation loop, rather than applying it post-hoc, proved critical: it forced both algorithms to provision adequate charger capacity at demand bottlenecks, yielding operationally stable deployment plans under peak load conditions.

Formal performance evaluation using Hypervolume, Generational Distance, and Spread indicators established that NSGA-II consistently outperforms MOPSO for this binary combinatorial problem structure. NSGA-II achieves a Hypervolume of 0.8412, a Generational Distance of 0.0014, and a median queue waiting time of 3.9 minutes across its Pareto front — compared to MOPSO values of 0.7205, 0.0039, and 52.2 minutes respectively. The queue waiting time distribution provided the most decisive evidence: the majority of MOPSO solutions violated the service-quality threshold, while NSGA-II solutions were concentrated well below the 20-minute user-tolerance boundary. The recommended NSGA-II optimal solution deploys 24 stations, achieves 86.4% city-wide spatial demand coverage, maintains a mean queue waiting time of 4.8 minutes, and delivers a net annual ROI of 14.2% for a total capital outlay of 2.14 Crore — confirming the commercial viability of public-private partnership EVCS deployment in Tier-1 Indian cities without perpetual state subsidy.

Sensitivity analysis confirmed robustness to EV growth rate, electricity tariff, and charger technology variations, while scalability benchmarks demonstrated polynomial runtime scaling for both algorithms, validating the framework's applicability to larger metropolitan deployments. Overall, this study demonstrates that rigorous multi-objective evolutionary optimization, combined with realistic stochastic demand modelling and analytical queueing constraints, can produce actionable, evidence-based EVCS deployment strategies that simultaneously satisfy spatial, operational, and financial requirements.

6.2 Future Work

Several extensions can further enhance the scope, accuracy, and practical utility of the proposed framework:

- **Road-network routing:** The current framework computes spatial coverage using Euclidean distances. Integrating OpenStreetMap-based road-network routing via graph-search APIs (e.g., OSRM) would produce topologically accurate service radii, particularly in areas with low road density or irregular urban form.

- **Heterogeneous charger-type optimisation:** The present model assumes a fixed AC charger blend per station. Extending the decision space to jointly optimise the ratio of AC slow chargers (7.4 kW) and DC fast chargers (50–150 kW) at each location would more accurately reflect real-world deployment decisions and their impact on queueing performance and capital cost.

- **Battery swapping station integration:** For the Indian two-wheeler and three-wheeler market — the dominant EV segments — expanding the optimization scope to include Battery Swapping Stations alongside conventional plug-in chargers would significantly improve the framework's coverage of actual urban charging demand.

- **Dynamic and real-time pricing:** The current ToU pricing structure is applied exogenously based on time-of-day patterns. Future work could incorporate real-time dynamic pricing responsive to localized substation load levels and live demand, potentially modelled through a Stackelberg game between the network operator and EV users.

- **Discrete-event simulation validation:** A discrete-event simulation engine would validate the steady-state M/M/c/K analytical results and model user behaviours not captured by Markovian assumptions, such as balking, reneging, impatience, and re-routing to alternative stations.

- **Statistical significance testing:** Incorporating Wilcoxon rank-sum or Mann–Whitney tests across multiple independent runs would formally confirm that NSGA-II hypervolume superiority over MOPSO is statistically significant, strengthening the comparative evaluation to publication standard.

- **GIS-based deployment tool:** Building an interactive GIS interface would allow municipal planners and infrastructure investors to explore the Pareto front visually, filter solutions by budget or coverage threshold, and export deployment-ready station specifications for field implementation.

- **Research publication:** Based on the results presented in this thesis, a full research paper will be prepared and submitted to a peer-reviewed journal or international conference in smart cities, transportation electrification, or evolutionary computation, documenting the four-objective formulation, NSGA-II vs. MOPSO comparison, and Indore case study findings.

These extensions can transform the current framework into a full-scale EV infrastructure intelligence platform, supporting sustainable, data-driven mobility planning for the rapidly electrifying urban ecosystems of Indian and other emerging-market cities.

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